

Figure 3: Secure boot chain of trust, as used by bootloaders such as MCUboot (Source: Original diagram by the authors, based on the MCUboot documentation. Reference: *https://docs.mcuboot.com*|)

from day one, since a device that cannot be patched will eventually become a liability.

It also helps to match the tools to the device. A tiny sensor with a few hundred kilobytes of memory may pair Zephyr with Mbed TLS and MCUboot, while a Linux gateway can run OpenSSL and Mosquitto comfortably. Starting from the NIST capabilities and checking each one against the design is a good way to avoid gaps.

Limitations and challenges

Open source is not a magic fix. Constrained devices have little memory and power, so heavy cryptography can drain batteries or slow responses. Provisioning keys and certificates to thousands of devices securely, and revoking them when something goes wrong, is difficult and often underestimated. Physical attacks and side channels can leak keys from unprotected hardware. Looking ahead, future quantum computers may weaken today's public-key algorithms, so long-lived devices should be built with the ability to upgrade their cryptography. Last, open source projects depend on maintainers, so it pays to choose actively supported ones.

Securing IoT devices comes down to knowing who is talking and keeping their conversation private. Open source frameworks make both achievable: libraries like Mbed TLS and OpenSSL supply TLS and DTLS, brokers like Mosquitto enforce certificate checks, and tools like MCUboot and EdgeX protect firmware and secrets. The best results come from combining them in layers, configuring them carefully and keeping them up to date. For a student or developer starting out, a good first project is a small device that connects to a Mosquitto broker using mutual TLS, because it covers identity, encryption and key handling in one go. END

References

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